# ECON 409: Final Project Guidelines

For your final project, you will develop your own forecasting strategy. This process will involve acquiring the necessary data, designing your strategy, and evaluating its performance through a historical back test. Over the following weeks, you will submit various project components at designated checkpoints, which will collectively account for 38% of your final project grade. Note that you will not be graded on the actual performance of your strategy; instead, your grade will be based on a final report that demonstrates your competence in meeting the project steps and includes a brief discussion of your results.

**Project Steps:**

1. Select the asset your team will forecast from the attached list of assets.
2. Procure the data required to develop your forecasting strategy. Some examples from class include:
    a. Inflation & Industrial Production → Currencies
    b. 12-month forward EPS → SP500 Sectors
    c. Google Trends Data on Inflation and/or employment → Currencies
    d. Sentiment Index → Currencies, SP500 or SP500 sectors

    Note that you may not just use the price of your asset to make forecasts (no pure momentum forecasting strategies). Therefore, you must pull at least two series: the historical prices of the asset you will be forecasting and a fundamental that will be used to help generate forecasts. *One of these series must be pulled from the Bloomberg Terminal.*
3. Describe your conceptual framework. Here you will explain the theory behind how your strategy works. You must explain why you chose each of the fundamental(s) in your forecasting strategy and how they relate to the price of the asset you are forecasting.
    a. For example, describing contrarianism works and why the sentiment of news articles containing terms related to gold may be a good contrarian indicator for the price of gold.
4. Describe your forecasting strategy. This must include when you enter and exit positions, and any data transformations applied. You should have both (i) an intuitive description of your strategy (using words) and (ii) a formal description that includes formulas.
5. Partition your data into a train and test set. You should have a *minimum* 7 years of data in your *testing* set.
6. Use the training set to produce a heatmap of your hyperparameters with the return of the strategy over the training set as the evaluation metric.
7. On the training set, visualize how your strategy works. Some examples of how to do this could be:

a. Zoom in to a given time period and show where your fundamental crossed above a filter then entered a position and where you subsequently exited. Mark these points on the fundamental and price series.
b. On your fundamentals graph show a point where your estimation strategy made a forecast and mark where this forecast was made on your fundamental(s) and price series.

8. Generate a back test of your forecasting strategy over the test set using walk-forward optimization. You should reoptimize your hyperparameters at least once per year in your back test.
    a. Visualize the values that each of the hyperparameters takes over the back test period
9. Plot your forecast signals over your fundamental(s) and the price series for your asset over the back test period.
10. Calculate a comprehensive set of performance metrics over the back test period for your forecasting strategy. Use the SP500 as your benchmark and Market Yield on U.S. Treasury Securities at 1-Year Constant Maturity for your risk-free rate. The metrics must include:

| | | |
|---|---|---|
| Annualized Return | Alpha | Sharpe Ratio |
| CCROR | Beta | Sortino Ratio |
| Max Drawdown | Up Alpha | Return on Account |
| Skewness of returns | Up Beta | Highest and lowest Month RoRs |
| Gini Coefficeint | Down Alpha | Weighted Directional Test (and interpretation) |
| Annualized Standard Deviation | Down Beta | Binomial Forecast Accuracy test (and interpretation) |

Calculate the same set of metrics for a strategy where you simply go long on the asset for the entire testing period. *(Note: Do not calculate the last two statistical tests — the Weighted Directional Test and the Binomial Forecast Accuracy Test — for the buy-and-hold strategy, as these are not applicable.)*

**Final Report Instructions:**

Once you have completed all project requirements, compile a final report that documents your entire project. The report must be a PDF. It should be well-organized, professionally presented, with clear figures and language, and include the following sections:

1. **Introduction.**

Summary of the question asked, method used, results.

2. **Strategy and Conceptual Framework**

Provide a comprehensive explanation of the underlying theory and rationale behind your approach. Detail your forecasting strategy by describing both the intuitive narrative and the formal elements (using formulas or pseudocode) that define how your strategy operates.

3. **Methodology**

Describe in detail the steps you took to implement your strategy. Explain your overall approach, outline the data sources used (including any preprocessing or transformations applied), and discuss the modeling techniques employed.

4. **Results**

Present the outcomes of your analysis by incorporating all visualizations from the project steps—such as strategy performance charts, hyperparameter heat maps, forecast signal plots, and hyperparameter trend charts. Include a table summarizing the performance metrics specified in Requirement 10, along with interpretations for the forecast accuracy tests and other metrics.

5. **Conclusions**

Interpret your strategy's performance by comparing it to both the S&P500 benchmark and a theoretical buy-and-hold strategy. Discuss any insights you gained, note the limitations of your approach, and propose potential improvements or directions for future work.

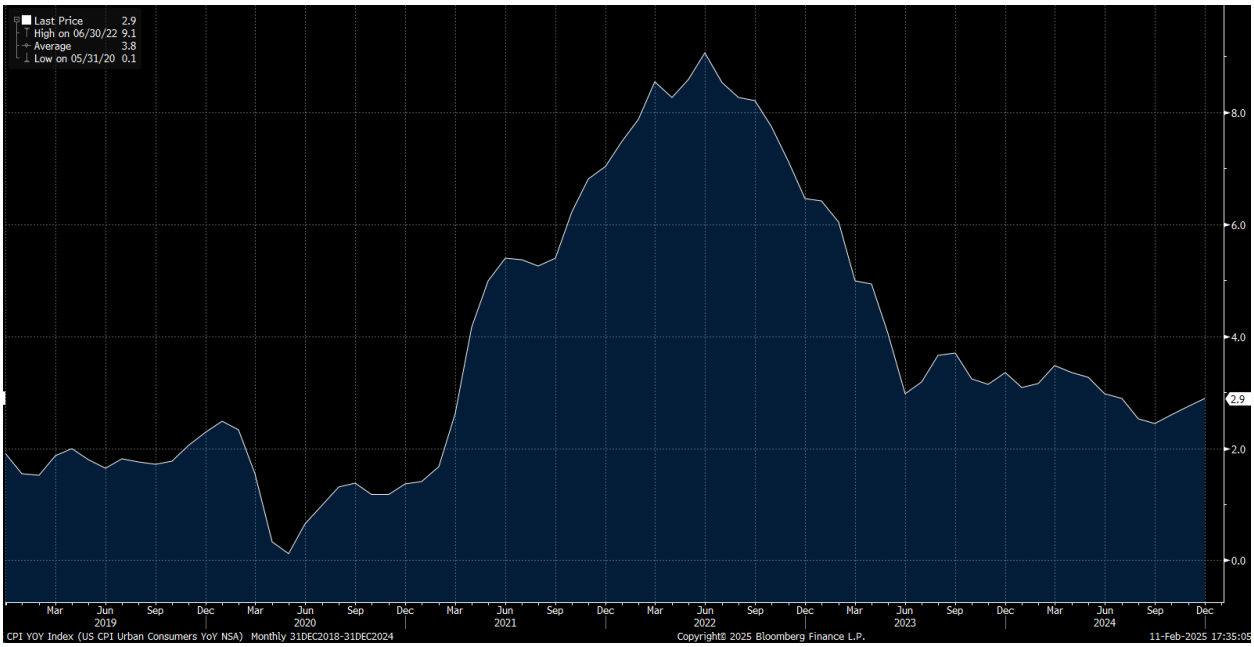
The final report should ***not*** contain any code.

**List of Assets**

**Currencies:** Foreign Exchange rates – United States Dollar, British Pound, Euro, Australian Dollar, Canadian Dollar, Swiss Franc, Mexican Peso, Japanese Yen (choose a pair of these currencies)

**Equities:** MSFT (Microsoft), JPM (J.P. Morgan and Chase), UPS, F (Ford Motor Company), XOM (Exxon Mobil Corporation), or any SP 500 Sector.

**TIMELINE:**

| **Checkpoint Date** | **Checkpoint Requirement** |
|---|---|
| End of Week 7 (Friday, 20 Feb) | Asset class choice to be submitted |
| End of Week 8 (Friday, 27 Feb) | A short draft describing your conceptual framework and an intuitive explanation of your forecasting strategy. A chart copied from Bloomberg showing at least one series of data you pulled for use in your strategy. |
| End of Week 9 (Friday, 6 March) | Heat maps of your hyperparameters on training data. |
| End of Finals Week (Thursday, 19 March) | Final Report Submission |

Note that for week 8 you must submit an up-to-date chart of one of your series pulled directly from the Bloomberg Terminal. You can take a screenshot or click "copy chart to clipboard" to get the chart. It should look something like this: